

HARTZ MOUNTAINS, AustraliaWMO: **949770**Lat: **43.201S**Lon: **146.768E**Elev: **831**StdP: **91.73**Time Zone: **10.00 (AUS)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.7	-1.2	-4.1	2.9	5.3	-2.5	3.4	1.5	20.8	3.7	18.1	2.8	9.2	250	0.892

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	24.3	15.5	21.7	14.5	19.6	13.7	16.4	21.8	15.3	20.3	14.3	18.6	6.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.4	17.9	13.4	10.6	17.0	12.4	9.9	15.9	49.1	21.7	45.8	20.4	42.8	18.8	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.9	14.4	-2.8	29.6	0.7	2.3	-3.2	31.3	-3.6	32.6	-4.0	33.9	-4.5	35.6		
(5)			-2.9	18.5	0.6	0.7	-3.3	19.0	-3.7	19.4	-4.1	19.8	-4.5	20.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	11.5	11.3	9.8	7.5	5.6	4.2	3.3	3.4	4.5	6.1	8.2	9.3
	DBStd	4.42	4.16	4.17	4.00	3.22	2.63	2.18	2.05	2.57	2.86	3.60	4.17	4.11
	HDD10.0	1376	29	31	53	89	140	176	208	203	166	132	85	64
	HDD18.3	4140	217	199	267	326	396	426	466	462	414	379	307	282
	CDD10.0	292	76	68	47	14	3	0	0	1	2	11	29	42
	CDD18.3	13	4	3	2	0	0	0	0	0	0	0	1	2
	CDH23.3	108	47	26	12	0	0	0	0	0	0	1	7	15
	CDH26.7	20	10	4	3	0	0	0	0	0	0	0	1	3
(14) Wind	WSAvg	7.5	7.1	6.4	6.8	7.3	8.2	8.0	7.9	8.0	8.3	7.8	7.4	7.2
(15) Precipitation	PrecAvg	999	63	56	63	71	83	91	100	122	101	96	79	74
	PrecMax	1288	149	168	127	193	231	184	197	220	199	197	140	148
	PrecMin	802	11	18	21	24	19	32	41	35	33	29	15	15
	PrecStd	128	36	37	29	35	50	40	42	47	46	39	31	30
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	26.8	24.7	18.4	14.2	11.3	10.0	12.6	15.3	20.7	24.2	25.4
		MCWB	16.3	16.9	15.4	12.3	9.7	8.8	6.9	8.6	10.0	12.0	14.0	15.2
	2%	DB	24.2	23.0	20.6	16.1	12.2	9.4	8.0	10.2	12.3	17.0	20.6	21.4
		MCWB	15.7	15.6	14.2	11.6	9.2	7.2	6.3	7.5	8.3	10.8	13.2	13.9
	5%	DB	21.2	20.4	18.0	14.1	10.5	8.1	6.9	8.4	10.5	14.5	17.5	18.6
		MCWB	14.5	14.7	13.2	10.5	8.6	6.5	5.5	6.4	7.5	9.9	11.9	13.0
	10%	DB	18.6	18.0	15.8	12.3	9.3	7.0	6.1	7.0	8.8	11.9	15.0	16.0
		MCWB	13.4	13.5	12.5	9.9	7.9	5.9	5.1	5.5	6.7	8.8	10.8	12.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	17.9	16.5	13.4	11.1	9.9	7.8	9.2	10.8	13.5	15.6	16.4
		MCDB	24.0	23.9	21.9	16.4	12.7	10.3	8.8	11.5	14.1	17.7	21.4	21.7
	2%	WB	16.3	16.5	14.9	12.3	10.0	7.9	6.6	7.8	9.1	11.7	14.0	14.9
		MCDB	22.1	21.4	19.2	14.9	11.2	8.8	7.5	9.6	11.6	15.8	19.2	19.9
	5%	WB	15.0	15.2	13.8	11.1	9.0	6.9	5.9	6.8	7.9	10.4	12.5	13.6
		MCDB	20.7	19.4	17.2	13.5	10.3	7.5	6.6	8.3	9.9	13.9	16.8	17.9
	10%	WB	13.7	14.0	12.7	10.0	8.1	6.2	5.3	5.8	6.8	9.0	11.1	12.3
		MCDB	18.1	17.5	15.6	12.0	9.1	6.8	5.9	6.8	8.4	11.9	14.8	15.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	8.8	7.3	5.6	4.2	3.6	3.7	4.4	5.7	7.2	8.3	8.5
		MCDBR	14.2	12.7	11.1	8.4	6.0	5.0	4.9	6.8	8.8	11.7	13.2	13.1
	5% WB	MCWBR	7.2	6.8	6.5	5.4	4.3	3.8	3.8	4.7	6.0	7.0	7.5	7.5
		MCWBR	12.9	11.4	10.1	7.8	5.3	4.1	4.5	6.4	8.0	10.7	12.1	12.0
(40) Clear-Sky Solar Irradiance	taub		0.316	0.304	0.303	0.290	0.282	0.272	0.275	0.287	0.308	0.311	0.308	0.313
		taud	2.566	2.619	2.613	2.629	2.630	2.638	2.637	2.613	2.533	2.536	2.570	2.564
	Ebn at Noon		1010	996	952	895	831	805	831	885	929	977	1013	1019
		Edh at Noon	104	93	85	71	59	53	57	71	90	100	102	105
(44) All-Sky Solar Radiation	RadAvg	6.23	5.35	3.88	2.60	1.57	1.25	1.44	2.15	3.26	4.67	5.70	6.23	
	RadStd	0.45	0.39	0.23	0.15	0.12	0.11	0.09	0.16	0.16	0.26	0.43	0.38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.33	N/A	N/A	+0.67	N/A	N/A	-100	+106	+14

Nomenclature: See separate page